

Liam Longfield

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Education

University of Texas at Austin, Cockrell School of Engineering (Expected Graduation May 2029)

Bachelor of Science in Mechanical Engineering | GPA: 4.0

Relevant Coursework: Mechanics of Solids, Dynamics, Thermodynamics, Statics, Intro to Programming (Python)

Experience

Longhorn Racing Electric FSAE - Suspension Team Member (September 2025 - Present)

- Designing the front suspension package, including the pushrod, rocker, damper, and anti-roll bar, to meet motion ratio targets, increase wheel rate through compression, and minimize compliance.
- Developed a first-principles Excel calculator to model steady-state vehicle characteristics—including motion ratios, ride frequencies, jacking forces, and roll rates—enabling rapid mapping, tuning, and validation of the lateral load transfer distribution.
- Designed double-wishbone suspension linkages and apexes for the 2026 FSAE car; used ANSYS FEA, buckling calculations, and a custom Python truss-analysis script to verify and optimize each linkage.
- Iterated a custom IPX7 electronics enclosure for ride-height sensors at all four corners of our vehicle; used Parker's O-Ring Handbook for groove sizing and O-ring selections.

Rugged Robotics - Engineering Intern (May 2026 - August 2026)

- Designed a desktop solvent-carrier ink fill station; integrated an ESP32 and load cell to dose with ± 2 mL accuracy.
- Iterated manifold face seals using Parker's O-Ring Handbook to isolate the solvent system from the environment.
- Modeled a 20-micron filter housing using surface modeling to seal the chassis and prevent dust ingress.
- Optimized and manufactured 3D-printed and machined jigs, with GD&T drawings, that increased robot-subsystem assembly efficiency by over 200% and enabled subsystem-level checkout and testing.

NASA Johnson Space Center - Engineering Intern (June 2025- August 2025)

- Engineered a fail-safe brake for NASA's Modular Robotic Vehicle (MRV) using a 220 lbf gas shock; prototype validation supported resumption of MRV rebuild efforts.
- Modeled and iterated in SolidWorks, using FEA to validate packaging and load capacity; created GD&T drawings for outsourced and in-house manufactured parts.
- Machined a fail-safe brake prototype from drawings on a manual lathe and milling machines with digital readouts, meeting positional tolerance requirements of ± 0.002 in.

FIRST Robotics Team 624 - President (August 2022 - May 2025)

- Led robot design, build, integration, sponsorship acquisition, and community outreach for an 80+ member team with 8 VPs; acquired and managed over \$20,000 in sponsorships and funding.
- Designed and iterated climb, end-effector, and deflector subsystems in Onshape and SolidWorks; helped the team place in the top 48 of more than 3,600 teams at the FIRST World Championship.

Skills

CAD/Analysis: SolidWorks, ANSYS, Lotus Shark; **Manufacturing:** GD&T, 3-axis CNC, manual mill, lathe.

Languages: French (B2 CEFR Proficiency)

Hobbies: Rock climbing, mountain biking, skiing, working on cars, hiking, and cooking